Modelling of a Galactic Flyby and of Stellar Formation

Anvel (Marguerite) de Sainte Marie d'Agneaux

1 Introduction

Two galaxies passing by each other is an incredibly complex event. The gravitational pull of one galaxy on the other, and the pull of the components of the galaxies between themselves, create complex structures such as tidal streams and send both galaxies into new directions. Due to the length of the event, they are also impossible to observe in their entirety. More complex simulations also allow the rewinding of one particular event to understand what led to a known object. Similarly, due to the size and complexity of the event, analytical solutions which explain the evolution of the objects accurately are also impossible. However, it is possible to create simple models which can run on any machine and are sufficiently accurate for most needs. Modelling them is thus the best way to understand how they evolve before, during and after the flyby. Galactic flybys in particular can be modelled with a simple algorithm involving only the gravitational pull from particle interactions. While this model will not reproduce small scale or particular events, it shows the broader evolution of galaxies after a flyby.

On the other hand, because it happens on a much smaller scale, the formation of a single star necessitates a more complex modelling. Stars form when some regions in molecular clouds become dense enough and collapse on themselves. Due to the velocity of the particles falling inward, if only the gravitational pull is taken into account, the sphere will collapse and expand in turn, indefinitely. To accurately model a star forming, other mechanisms must be taken into account. Two such mechanisms are the internal pressure and the viscosity of the collapsing gas which apply to each particle due to the neighbouring ones.

This project is composed of two parts. In the first part, we will implement a numerical model of a moving lower mass galaxy flying by a stationary higher mass galaxy. In the second part, we will attempt to model the formation of a star. Both models use N-body simulations. Due to its smaller size, the stellar model is of much higher resolution, as the same number of particle can be used to model the large galactic fly-by and the much smaller star formation. However, due to coding mistakes, the stellar formation simulation did not work, and thus it is only mentioned in theory in this report.

2 Implementation

Both models were based on the same structure. A main file called forward multiple functions to initiate, evolve and plot the galaxies and stars.

The first set of functions placed N particles in a circle around a centre of mass (x, y) with a normal distribution for the radius, and a uniform distribution for the angle. It calculated an initial velocity in the x and y direction using 0.4 of a Keplerian velocity for the galaxies, which is dependent on the radius from the centre of mass, and was taken as null for the stars. Then the acceleration of each particle was calculated from this initial velocity and the position of all the other particles in the simulation. A second set of functions updated the positions,

velocity and acceleration of the particles after a time step dt using a leapfrog algorithm. It first calculated an updated velocity after half a step using the previous acceleration, then calculated an updated position using this new velocity, before getting the updated acceleration and taking the second half-step in velocity using this new acceleration. This is the leap-frog algorithm:

```
new_v = v + dt/2 * acc
new_p = p + dt * new_v
new_acc = getAcceleration(*args)
new_new_v = new_v + dt/2 * new_acc
```

Where getAcceleration() calculates the acceleration of the particles using the different equations discussed below. A last set of functions created and updated the plots, making a series of image every few time steps.

2.1 Equation of motion

The equation of motion was the most complex part of this model, as it had to be calculated individually for each particle. It was present in two different ways between the two models, and this was the largest difference between them. One of the model only included gravitational effects on the motion of the particles, while the other one included both the gravitational effects, the pressure gradient, and the effects of viscosity. The gravitational effects were a weighted sum of the pull of the other particles as a function of their mass and distance to the particle of interest:

$$\vec{F_j} = m_j G \sum_{i \neq j} \frac{m_i}{r_{ji}^3} \vec{r_{ij}} = m_j \vec{a_j} \Leftrightarrow \vec{a_j} = G \sum_{i \neq j} \frac{m_i}{r_{ji}^3} \vec{r_{ij}}.$$

The pressure gradient was slightly more complex, as it needed to be calculated as a function of the average density around the particle of interest instead of its interaction with the other individual particles. For this, we define a density kernel at each point in space which depends on the presence of other particles within the vicinity. The density at a location r is:

$$\rho(\vec{r}) = \sum_i m_i W_i(|\vec{r} - \vec{r_i}|, h).$$

Where W_i is a smoothing kernel, such that:

$$W_i(q,h) = C \begin{cases} (2-q)^3 - 4(1-q^3) & \text{if } 1 > q \geq 0, \\ (2-q)^3 & \text{if } 2 > q \geq 1, \\ 0 & \text{if } q \geq 2. \end{cases} \quad (1)$$

And $C = \frac{5}{14\pi h^2}$, $q_i = \frac{|\vec{r} - \vec{r_i}|}{h}$ and h is a constant which we chose to be $0.02 \cdot R_\odot$. From this, assuming an isothermal equation of state, it is possible to obtain the pressure : $P = \kappa \rho^2$, and from it, the pressure gradient:

$$\vec{\nabla} P(\vec{r_j}) = 2\kappa \sum_i m_i \vec{\nabla}_j W(|\vec{r_j} - \vec{r_i}|, h)$$

Finally, the impact of viscosity is simply a constant ν multiplied by the speed of the particle j at time t. The exact values used in this report for the different models can be found in Appendix A.

2.2 Acceleration of the code

As an N-body simulation is extremely taxing for the machine running it, multiple methods were used to accelerate the process, and make it less taxing. There were two main possible paths. The first one was to use the Numba package and compile the Python code into a language which the machine has less difficulty running. It also included the parallelisation of some parts of the code. In particular, for loops are extremely slow in Python. Running multiple of the items parallelly accelerated the code significantly. The second method was to entirely erase the need for for loops by using NumPy array maths.

The code which removes the need for loops is reproduced here:

```
p_sub_0 = np.repeat(p[0,:], len(p[0,:]), axis=0).reshape(len(p[0,:]),
                            len(p[0,:]))
p_sub_1 = np.repeat(p[1,:], len(p[1,:]), axis=0).reshape(len(p[1,:]),
                            len(p[1,:]))
r_ji_x = p[0,:].transpose()-p_sub_0
r_ji_y = p[1,:].transpose()-p_sub_1

#Clip anything too close to 0
r_ji = np.clip(np.sqrt((r_ji_x)**2 + (r_ji_y)**2), 0.1*Rg, None)

#Get the acceleration
acc[0,:] = - G * sum((mp[:] / (r_ji)**3)*r_ji_x)
acc[1,:] = - G * sum((mp[:] / (r_ji)**3)*r_ji_y)
```

It creates a larger matrix of size *number of particle* $\times$ *number of particles*, such that when calculating the resulting radius, it should produce this sequence of arrays:

$$r_{ji,x} = \begin{bmatrix} p_{x1} \\ p_{x2} \\ p_{x3} \end{bmatrix} - \begin{bmatrix} p_{x1} & p_{x2} & p_{x3} \\ p_{x1} & p_{x2} & p_{x3} \\ p_{x1} & p_{x2} & p_{x3} \end{bmatrix} = \begin{bmatrix} p_{x1} - p_{x1} & p_{x1} - p_{x2} & p_{x1} - p_{x3} \\ p_{x2} - p_{x1} & p_{x2} - p_{x2} & p_{x2} - p_{x3} \\ p_{x3} - p_{x1} & p_{x3} - p_{x2} & p_{x3} - p_{x3} \end{bmatrix}$$

Then:

$$r_{ji} = \sqrt{\begin{bmatrix} p_{x1} - p_{x1} & p_{x1} - p_{x2} & p_{x1} - p_{x3} \\ p_{x2} - p_{x1} & p_{x2} - p_{x2} & p_{x2} - p_{x3} \\ p_{x3} - p_{x1} & p_{x3} - p_{x2} & p_{x3} - p_{x3} \end{bmatrix}^2 + \begin{bmatrix} p_{y1} - p_{y1} & p_{y1} - p_{y2} & p_{y1} - p_{y3} \\ p_{y2} - p_{y1} & p_{y2} - p_{y2} & p_{y2} - p_{y3} \\ p_{y3} - p_{y1} & p_{y3} - p_{y2} & p_{y3} - p_{y3} \end{bmatrix}^2}$$

Which produces an n by n matrix where each row is the radius between a particle and all the others. Finally, the calculation of the acceleration sums over the entire row, summing over the contribution of each particle i on the acceleration of particle j.

To test the comparative speed of the three methods, I used a smaller simulation, with only 500 particles in one galaxy, and 250 in the other. The three methods being: with no acceleration, using Numba, and using array maths. The fastest was the Numba method, at 529.2 seconds, followed by the array method at 1309 seconds, more than double the time. As expected, the method with no acceleration was last, taking nearly 1500 seconds. So, while compiling the code in Numba and running multiple loops in parallel is the best option, reducing the number of for loop did significantly improve the running of the simulation. The large matrix it has to build and the plotting end up being the longer operation instead of the for loops, and they take less time.

There are probably ways in which the no loops methods could be improved, but due to time constrains, it could only be tested with basic mechanisms. It does work as a proof of concept.

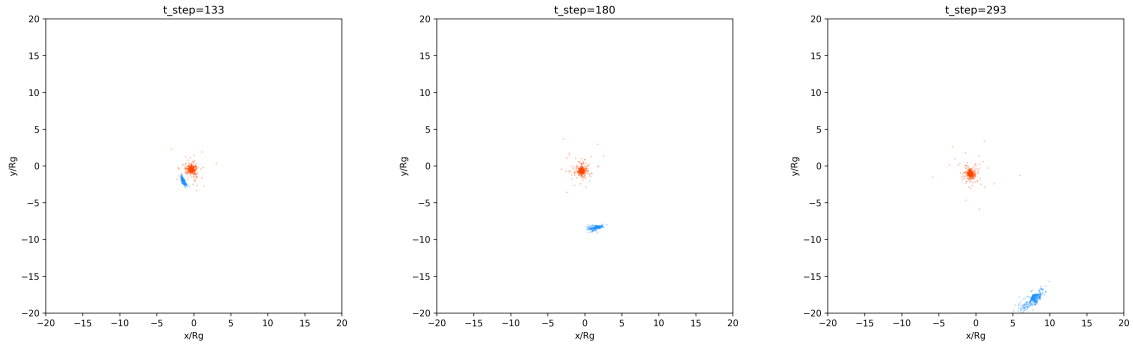

Figure 1: Evolution of the fly-by of an extremely low mass galaxy next to a Milky Way sized galaxy.

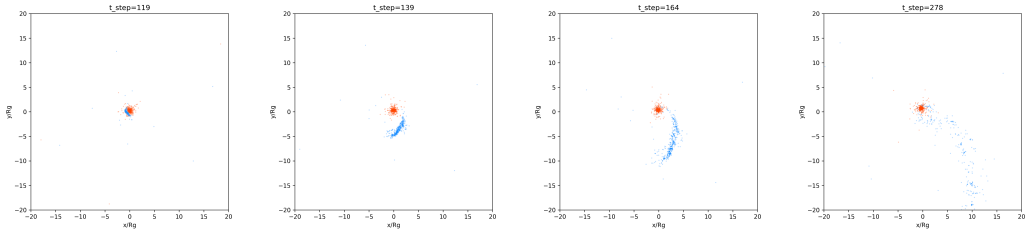

Figure 2: Evolution of the flyby of a lower mass galaxy next to a Milky-Way sized galaxy.

3 Results

3.1 Galaxy Merger

Using the galaxy merger model, I run four different simulations. The parameters used for all of them can be found in Appendix A. The first one had a Milky Way size galaxy interacting with an extremely low mass galaxy, of only 10^5 times the mass of the Sun. The second was interacting with a larger galaxy, although still low mass, of about 10^{10} the mass of the Sun. The third also included a change in mass, with two galaxies of about equal mass interacting. The last was slightly different in that it included the Milky Way size galaxy and the $10^{10} M_\odot$ ones, but with a head-on collision instead of the offset fly-by which the others modelled.

As we can see in figure 1, the extremely low mass galaxy, which is arriving from the right, slightly above the other galaxy, gets slingshotted around the higher mass one. Contrary to what is expected from higher mass galaxies, it does not form a tidal stream, but instead spreads across a half circle and continues on a new, nearly perpendicular trajectory. It seems that if the simulation continued, the lower mass galaxy would become unbound and its individual particle would get ejected along their previous trajectory, in a half circle. This is probably due to the fact that the galaxy is too small and fast to be slowed down by the gravitational pull of the larger galaxy. While the galaxy does slingshot it, no particles trail behind because they are too small and fast to be impacted. However, it does break apart because the galaxy itself is not strongly gravitationally bound because it is too small.

As we can see in figure 2, the lower mass galaxy is also slingshotted by the high mass one when the mass ratio between the two is smaller. However, the lower mass galaxy is still large enough for part of its particles to get pulled into the higher mass one. A tidal stream forms, and while some particles continue in their trajectory, some remain in the vicinity of the non-moving galaxy. They are absorbed into it after a time.

In figure 3, we can see that the equal mass galaxy interaction is very different. While the

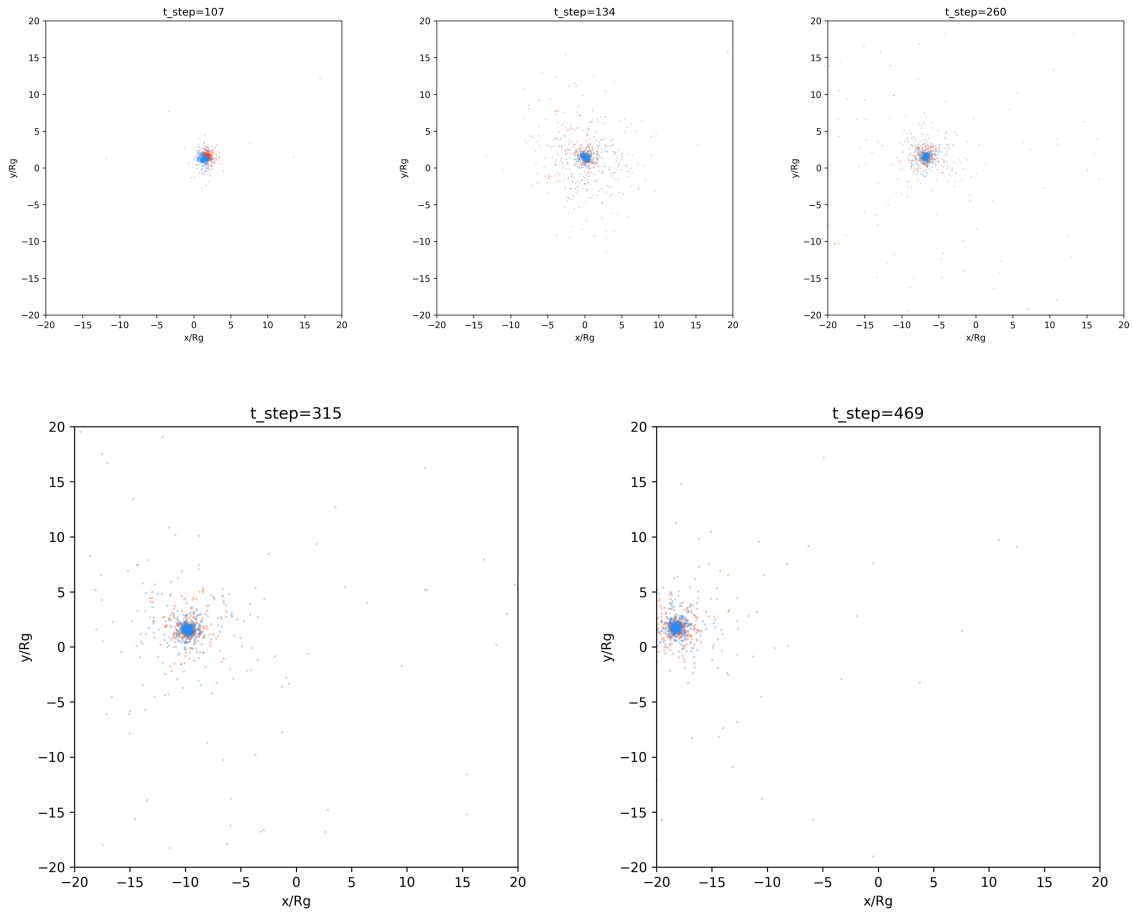

Figure 3: Evolution of the flyby of two galaxies of about equal size, both of the size of the Milky Way.

moving galaxy originally starts orbiting around the non-moving one, as is visible in the first panel, it very quickly falls into it. Particles from both galaxies get ejected into higher orbit. Part of the motion of the second galaxy is transferred to the first, such that both galaxies start moving together in the direction the second one was going toward. Ultimately, both galaxies will probably merge into one, moving at a speed depending on the combined speed of both of them when they merged.

Finally, the last simulation can be seen in figure 4. In this simulation, a lower mass galaxy collides head on with a higher mass one. While the interaction begins in much the same way as the similar mass ration interaction with an offset, it does not continue similarly. The lower mass galaxy is slingshotted into a different direction, and is ejected into an arc which resemble more the extremely low mass galaxy interaction. This is however where the resemblance stops, as this arc is dispersed into a cloud of unbound particles. It seems that a higher proportion of them are absorbed into the non-moving galaxy, but on a higher orbit than in the other simulation with a similar mass ratio.

3.2 Stellar Formation

A large set of simulations were run, with varying radii and masses, which in turn also varied the smoothing factor as it is directly dependent on the radius of the star. It also modified the minimum value the distance between particles could take within the equation for the gravitational acceleration as this was taken as $0.1R_{star}$. In total, 14 simulations were run,

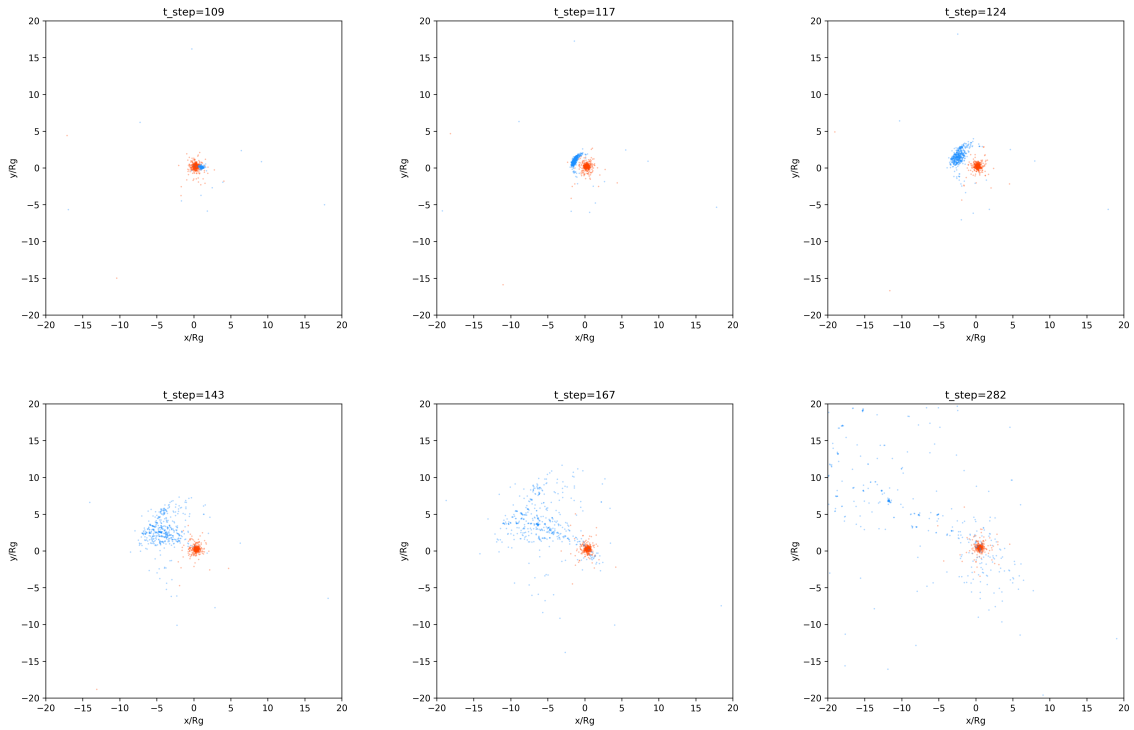

Figure 4: Evolution of a lower mass galaxy colliding head on with a galaxy of Milky-Way size.

which are summarised in Appendix A. The masses taken ranged from $0.1M_\odot$ to $M_\odot$, and the radii from $0.1R_\odot$ to $R_\odot$. In addition, one simulation was run with $0.3R_\odot$ and $M_\odot$ but a smoothing factor of $0.1R_{star}$ instead of $0.02R_{star}$. The last was a similar simulation, with $h = 0.02R_{star}$ once again, but with two 2D gaussians instead of only one at $t = 0$.

First, we can look at the evolution of a single simulation across time in this model. It is shown in figure 5. This simulation is of 2000 particles with a total mass of $M_\odot$ and a radius of $0.3R_\odot$. The smoothing factor was the most common one at $0.02R_{star}$.

A first point of interest is that for a same mass, stars of decreasing radius appeared larger in the plots (see fig. 6). These are simulations for stars of the mass of the Sun, and radii of $R_\odot$ (left), $0.7R_\odot$ (middle) and $0.3R_\odot$ (right). While it was run, the simulation of the same mass with a radius of $0.1R_\odot$ is not shown here because the particles were ejected within 50 time steps as they formed too close to each others.

4 Discussion

As mentioned in the Introduction, numerical simulations are necessary to understand phenomenons which can not be observed in their entirety and are too complex for analytical solutions. Although quite basic, the simulations in this report already represent the phenomenons they model quite well.

Concerning their accuracy, the results within the galaxy merger simulation depend heavily on the initial radial velocity of the particles. Giving them too much velocity at first will make them fly away instead of rotating around the centre of mass of the galaxy. This means that the galaxies will disperse before they interact, or that the interaction can be impacted by this too high initial velocity. It is also an extremely simplified model, which only include the gravitational interaction of heavy collisionless particles. More complex models show that some collision are necessary to describe accurately the evolution of galaxies. Furthermore, due to

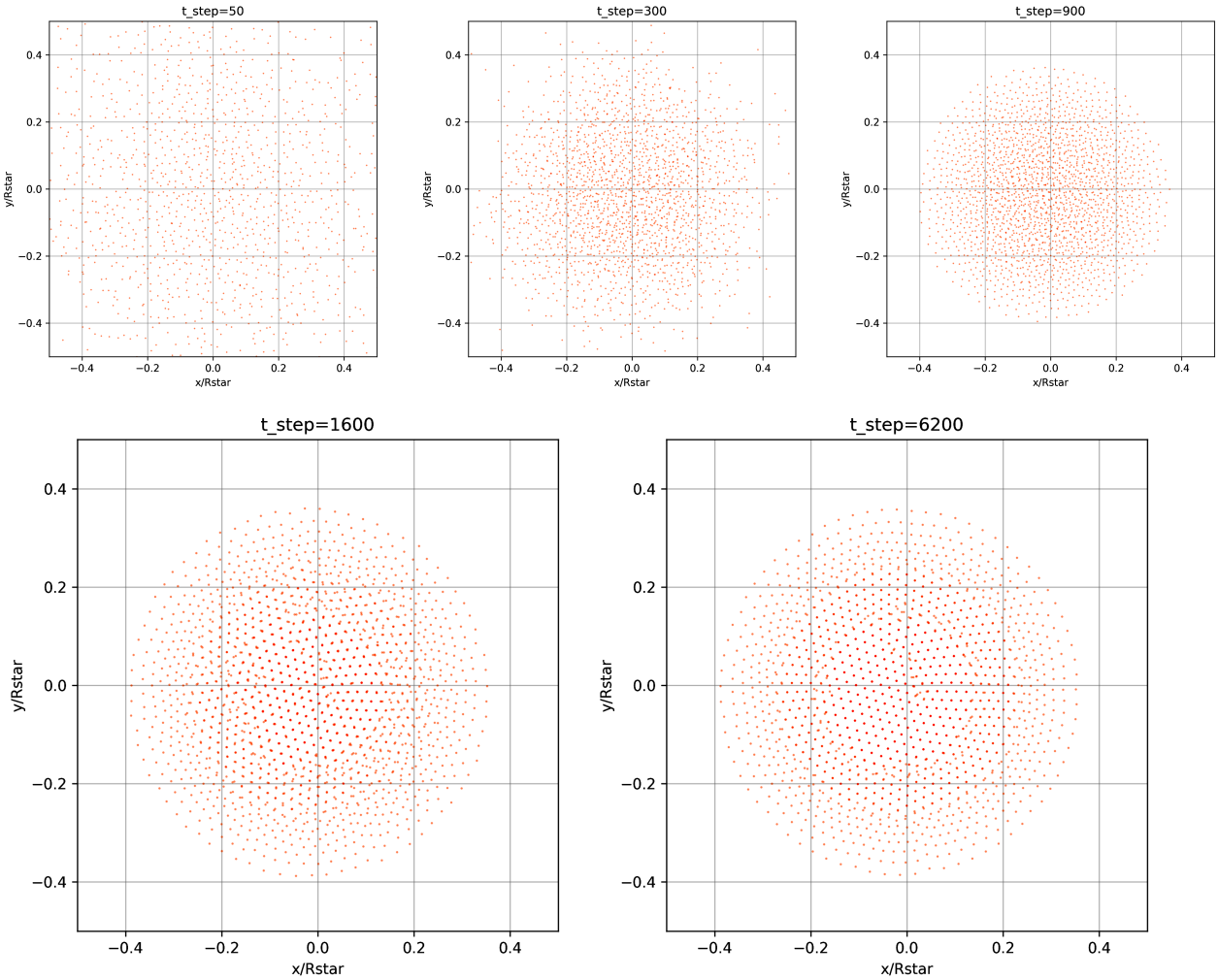
t_step=50
t_step=300
t_step=900
t_step=1600
t_step=6200
y/Rstar
x/Rstar
0.4
0.2
0.0
−0.2
−0.4

Figure 5

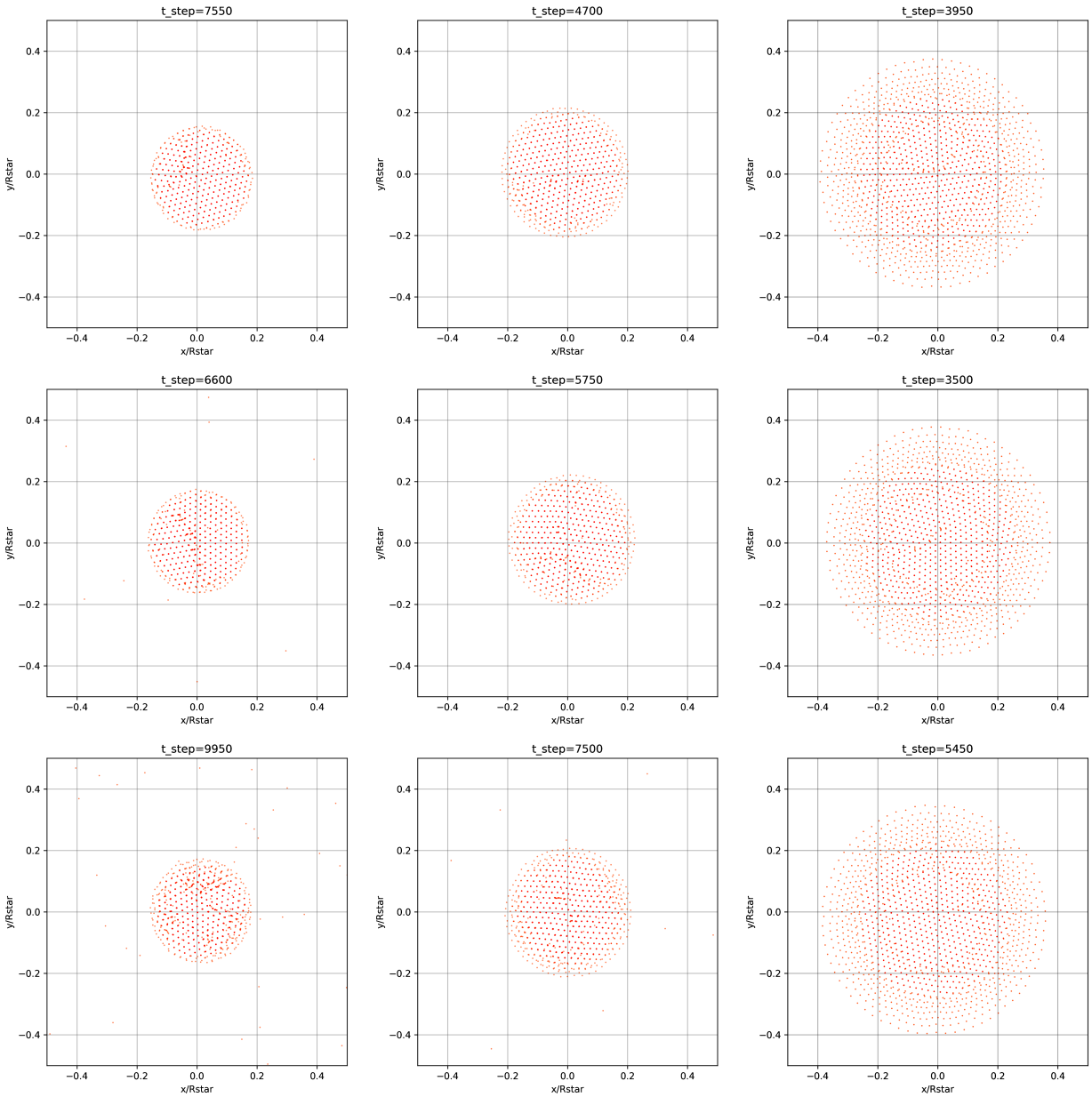
t_step=7550
t_step=4700
t_step=3950
t_step=6600
t_step=5750
t_step=3500
t_step=9950
t_step=7500
t_step=5450
y/Rstar
x/Rstar
0.4
0.2
0.0
−0.2
−0.4

Figure 6

the limit imposed by the machine it is run on, the simulations have to be low resolution. Each increase in the number of particles proportionally increases the time it takes the simulations to run, and the power it takes to run them. Most portable computers are uncapable of running more complex models. Using Numba or reducing and removing for loops and plotting routines do improve this point, but neither can improve it indefinitely.

While the stellar formation model produced with this report does not work, it is possible to make some assumptions as to what works and what could be improved. It is slightly more complex, and due to the smaller size of the object being modelled, it can be simulated in higher resolution by using the same number of particles as within the galaxy simulations. It does however make some assumptions which could impact the results. In particular, it does not take into account magnetic fields, which are known to be very strong in star formation discs and young stars. It also assumes an isothermal model, which does not take changes in temperature into account. Temperature is important in star formation, as it is one of the variables in the calculation of Jeans' mass. This mass determines whether a molecular cloud can collapse into a star or not.

Finally, both models are made in closed box simulations, where only the initial particles and objects interact. This is not an entirely accurate representation, especially for the stellar formation model. The galaxies in the galaxy merger model are large enough and far enough away from other heavy objects that they should not be overly impacted. However, star form in groups with other stars, and are surrounded by many other elements. It is mentioned in the instructions for the model that the final stars are heavier in the centre than they should be. This could be linked to this problem. In these closed box model, all the particles end up in the star after a certain time, instead of some of them escaping or being pulled away. Radiation pressure is also not taken into account. Particles should be pushed away from each others by their emissions.

In conclusion, these simulations both are accurate models of the astrophysical phenomena they aim to represent. However, multiple effects which have been shown to impact both the events were not included in the simulations. Thus, while they can be used to understand the movements of large structures, the minute details can not be understood with their help.

The code can be found at GitHub Repository, along with the LATex version of this report and the plots made during the simulations

5 Appendices

5.1 Appendix A

Common to all simulations	
M_g	$1.54 \cdot 10^{12}$
R_g	$9.5 \cdot 10^{17}$
τ	$\frac{R_\odot}{G \cdot M_g}^{0.5}$
dt	$0.01 \cdot \tau$
n_{dim}	2
n_{steps}	10 000
Simulations	
1 - Extremely Low Mass galaxy	
Galaxy 1	
N_1	1000
M_1	M_g
R_1	R_g
Offsets (position, velocity)	None
Galaxy 2	
N_2	500
M_2	$1.54 \cdot 10^5 \cdot M_\odot$
R_2	$3.8 \cdot 10^{17}$
$x_{0,2}$	$15 \cdot R_g$
$y_{0,2}$	$2.5 \cdot R_g$
$v_{x_0,2}$	$\frac{-R_g}{2 \cdot \tau}$
$v_{y_0,2}$	0
2 - Low Mass galaxy	
Galaxy 1	
N_1	1000
M_1	M_g
R_1	R_g
Offsets (position, velocity)	None
Galaxy 2	
N_2	500
M_2	$1.54 \cdot 10^10 \cdot M_\odot$
R_2	$3.8 \cdot 10^{17}$
$x_{0,2}$	$15 \cdot R_g$
$y_{0,2}$	$2.5 \cdot R_g$
$v_{x_0,2}$	$\frac{-R_g}{2 \cdot \tau}$
$v_{y_0,2}$	0

3 - Equal Mass galaxy	
Galaxy 1	
N_1	1000
M_1	M_g
R_1	R_g
Offsets (position, velocity)	None
Galaxy 2	
N_2	1000
M_2	M_g
R_2	R_g
$x_{0,2}$	$15 \cdot R_g$
$y_{0,2}$	$2.5 \cdot R_g$
$v_{x_0,2}$	$\frac{-R_g}{2 \cdot \tau}$
$v_{y_0,2}$	0
4 - Low Mass and Head On Collision	
Galaxy 1	
N_1	1000
M_1	M_g
R_1	R_g
Offsets (position, velocity)	None
Galaxy 2	
N_2	500
M_2	$1.54 \cdot 10^{10} \cdot M_\odot$
R_2	$3.8 \cdot 10^{17}$
$x_{0,2}$	$15 \cdot R_g$
$y_{0,2}$	0
$v_{x_0,2}$	$\frac{-R_g}{2 \cdot \tau}$
$v_{y_0,2}$	0

Figure 7: Table summarising the parameters of the Galaxy Merger Simulations